

Pressure-Flow Scour Due to Vertical Contraction: A Review

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Abstract: Climate and land-use/land-cover changes are amplifying hydrological uncertainties, which are increasing the probability of flood magnitudes exceeding their design values. Consequently, bridges are more likely to get submerged. The vertical contraction of flow during a submergence event causes the flow beneath the bridge to accelerate and transit from atmospheric flow to pressure flow. The pressure flow increases the velocity gradient and bed-shear stresses, thereby increasing the scouring potential of the flow and the risk of bridge failure. Laboratory and numerical experiments have been performed to relate the magnitude of pressure-flow scour to the flow geometry and flow, bed, and fluid properties. These developed prediction equations have found a place in design manuals. Nevertheless, many aspects of the pressure-flow scouring are confounding and are still unknown after 3 decades of research. In this article, an overview of the pressure-flow scour phenomenon is presented. The focus is on the important but special case of pressure flow with no pier that has received maximum attention in the literature. The experimental studies on pressure-flow scour are summarized, and inconsistencies and gaps in the reported results are highlighted. A compendium of data compiled from all existing data sets is used for evaluating the design equations and investigating their relative strengths and shortcomings. Based on the anomalies identified in the literature, the article questions the applicability of empirical and semitheoretical models for the pressure-flow scour phenomenon and suggests directions for future research. **DOI: 10.1061/(ASCE)HY.1943-7900.0001943.** © 2021 American Society of Civil Engineers.

Introduction

The single most important reason for a bridge failure is scouring of the bridge foundation (Gaudio et al. 2012). The total scour at a bridge crossing includes the long-term degradation of the bed as well as local and general scour (Richardson and Richardson 2008). General scour occurs because of the higher flow velocities due to flow contraction or a channel bend. The waterway may be contracted laterally and/or vertically at a bridge site. In comparison with horizontal contraction scouring, vertical contraction scouring, which occurs when a bridge gets submerged, leading to pressure-flow conditions, has received less attention.

In fact, existing bridges have all been designed for open-channel flow conditions (Carnacina et al. 2019b). The design discharge that must be passed through the contraction is calculated for some return period of the flood based on the importance of the bridge, but often without due consideration of climatic trends (Carnacina et al. 2019b). A study on climate change by Transportation Research Board (2008) has warned of increased risks in the magnitude of riverine and coastal flooding due to increased precipitation events, storm surges, sea level rise, and hurricane activity. Thus, there is an increasing possibility for existing bridges to face the conditions of vertical flow contraction in their lifetime. So, for the safety analysis of the existing bridges and future design guidelines, a comprehensive understanding of the pressure-flow scouring due to vertical contraction is necessary.

In vertical contraction, the water surface rises above the bottom chord of the bridge, and a pressure-flow situation is created in the waterway under the bridge (Lyn 2008; Umbrell et al. 1998). Fig. 1 shows the definition sketches of vertical contraction scouring for different cases reported in the literature: (1) bridges without girders and piers, (2) bridges with girders but without piers, and (3) elevation view of bridges with piers, respectively. The vertical contraction accelerates the flow in the passage, besides changing the nature of flow field in the contraction. The increased flow velocity might either initiate sediment motion on a previously uneroded bed or may add to the scour already present in the contraction (Lyn 2008). In more complex situations, the bridge may get completely submerged and a part of flow passes over the bridge, as in the weir flow in Fig. 1(b). In other instances, piers may also be present under the bridge [Fig. 1(c)], and the pressure flow due to vertical contraction may increase the pier scour as well (Abed 1992; Carnacina et al. 2019b).

However, the main focus of this article is on the special case of no pier (with and without weir overflow) because it has received most of the attention in the literature. In this article, the pressure-flow scour due to vertical contraction with no pier (for both submerged and unsubmerged cases) is denoted by y_p and its equilibrium value by y_{pe} . Unless otherwise specified, the term pressure-flow scour in this article will refer to y_{pe} . The more general case of pier scour under pressure-flow condition is briefly discussed in a subsequent section.

Like all other types of scouring, the pressure-flow scouring is dependent on the flow geometry and the flow, bed, and fluid properties. These independent variables are listed in Table 1. Besides, in many scour prediction formulations, a condition for the initiation of the sediment motion, either critical velocity, V_c , or the critical tractive stress, τ_{cr} , is included in the set of independent parameters. These critical parameters are in turn functions of the flow and bed properties. Thus, the problem of pressure-flow scouring due to vertical contraction can be stated as follows: given a set of independent parameters, estimate the magnitude of maximum equilibrium

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Note. This manuscript was published online on October 9, 2021. Discussion period open until March 9, 2022; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Hydraulic Engineering*, © ASCE, ISSN 0733-9429.

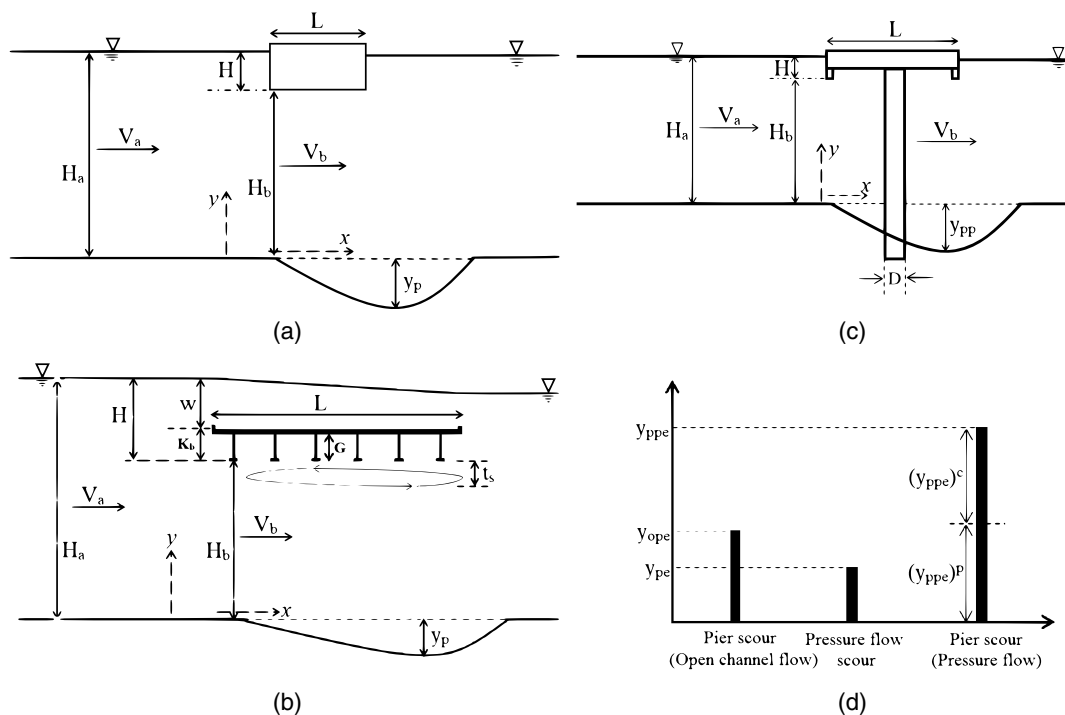


Fig. 1. Schematic of pressure-flow scouring due to vertical contraction: (a) simple vertical contraction; (b) vertical contraction with girders and having weir overflow; (c) elevation view; and (d) description of the pier scour in pressure-flow conditions.

Table 1. Factors affecting pressure-flow scour and pier scour in pressure-flow conditions

Category	Properties
Flow geometry	Channel width, B Pier diameter, D Contracted passage at the bridge site, H_b Length of the bridge, L Weir overflow, w
Flow properties	Approach flow depth, H_a Approach flow velocity, V_a
Bed properties	Median sediment size, D_{50} Specific gravity of sediment, S_g Gradation of the sediment, σ
Fluid properties	Viscosity of the fluid, ν Density of the fluid, ρ

pressure-flow scour, y_{pe} , geometry of the scour hole, and temporal evolution of the scour hole.

This article first reviews the experimental studies for addressing the pressure-flow scour problem. Next, the reliability analysis of the existing pressure-flow scour prediction models is carried out to shed light on their strengths and limitations. It is followed by a brief section on pier scour in pressure-flow conditions. Some critical questions and the importance of the phenomenon to the advancement to science and engineering practice are discussed next. The discussion is aimed at critically assessing the gaps in our understanding of the pressure-flow scouring process. Finally, the article concludes with a summary of what is known and what remains to be explored regarding pressure-flow scour. Some suggestions on the directions of future research are also proposed.

Experimental Studies on Pressure-Flow Scour

Dimensional Analysis

Pressure-flow scouring, like other types of scouring, is a complex process involving many variables, including those associated with the flow geometry and flow, bed, and fluid properties. A dimensional analysis of the independent parameters (Table 1) and pressure-flow scour, y_p , based on Arneson and Abt (1998), Kocyigit and Karakurt (2019), and Carnacina et al. (2019b) yields Eq. (1). [Eqs. (S1) and (S2) give more details].

$$\frac{y_p}{H_a} = f\left(\frac{H_b}{H_a}, \frac{V_a}{V_c}, \frac{w}{H_a}, \frac{D_{50}}{H_b}, \frac{t}{T}\right) \quad (1)$$

Ettema et al. (2011) categorized independent variables as primary and secondary variables. The primary variables that decide the magnitude of the potential maximum pressure-flow scour include H_b/H_a , w/H_a , and D_{50}/H_b . The secondary variables that usually lead to scour depths less than the potential maximum scour include V_a/V_c and t/T , where T is the timescale used to nondimensionalize the time, t . The relative importance of the dimensionless groups, listed in Eq. (1), are briefly outlined as follows:

- y_p/H_a represents the dimensionless scour depth at time t , which assumes the value y_{pe}/H_a at equilibrium. This group has been used as a dependent variable in the regression by Arneson (1997) and Lyn (2008) purely on empirical basis. Another related group, $(y_{pe} + H_b)/H_a$ denoting the ratio of the total clearance inside the bridge opening and the upstream flow depth at equilibrium, has been derived by Umbrell et al. (1998) using the continuity equation.
- H_b/H_a , the ratio of the bridge opening and the upstream flow depth, defines the geometry of the contraction along with the group w/H_a in the case when some part of the discharge passes

over the bridge as weir overflow. The sensitivity analysis of the dimensionless groups listed in Eq. (1) for the positive and unsubmerged scour cases is carried out and reported in the Supplemental Materials. It is clear that H_b/H_a is a dominant variable in the estimation of y_{pe}/H_a .

- The densimetric Froude number, F_a^* , defined as $V_a/\sqrt{g(S_g - 1)H_a}$, can be reduced to V_a/V_c (Carnacina et al. 2019b) and represents the upstream bed mobility. Most of the studies on pressure-flow scour have been clear-water studies ($V_a/V_c < 1$). Perhaps, the motivation has been the fact that the other forms of scour, especially pier scour, have the highest magnitude when $V_a/V_c \sim 1$ (Melville 2008). In fact, Melville (2014) reported that the pressure-flow scour attains first peak at $V_a = V_c$ and then decreases and attains a second peak (lower than the first peak) at $V_a = 2V_c$ and drew a similarity between the local pier scour in open-channel conditions and pressure-flow scour. Thus, clear-water experiments provide a conservative estimate of the pressure-flow scour even when mostly the actual field conditions are live bed and the V_a can be higher than V_c . In the sensitivity analysis reported in the Supplemental Materials, it is observed that V_a/V_c is a significant variable in the estimation of y_{pe}/H_a .
- D_{50}/H_b represents sediment coarseness ratio that has been reported to have no effect on y_{pe}/H_a (Arneson 1997; Arneson and Abt 1998). Arneson (1997) argued that the effect of D_{50} is indirectly incorporated in the V_c . Most of the later studies have not reported the direct effect of D_{50} . A possible explanation may be borrowed from the pier scouring where there is no effect of the D_{50} when D_{50}/D , D being the diameter of pier, is less than 0.02 (Melville 2008). In all the reported pressure-flow scour experiments, the ratio D_{50}/H_b is very small (<0.03). This point has also been highlighted by Hahn and Lyn (2010). In the sensitivity analysis also, it is found that D_{50}/H_b is the least significant variable in the estimation of y_{pe}/H_a .
- t/T represents the time factor of the evolution of the scour hole where T is the timescale. T is a function of the flow and sediment properties and is discussed in the next section.

The groups listed in Eq. (S2) but not listed in Eq. (1) are as follows:

- B/H_a , defined as the aspect ratio of the channel, is usually large. So, the effect of the width of channel, B , on y_p is neglected. The assumption is further strengthened by the fact that unlike close to pier or abutment, there is not any lateral obstruction to the flow in pure vertical contraction.
- The effect of length of the bridge, L , on y_p has not been reported in any study and hence is not considered in Eq. (1).
- σ is the sediment uniformity coefficient. The sediment is treated as uniform for $\sigma < 1.4$ (Dey and Debnath 2001). In the laboratory experiments reported in the literature, the sediment is uniform, and hence σ has not been included in any of the models. S_g is the specific gravity of the sediment. It is not considered further because its value is constant for all the experiments in the reported studies.
- The approach flow Reynold's number, $(V_a H_a)/\nu$, becomes significant when the flow regime is laminar or transitional (Carnacina et al. 2019b). Open-channel flows usually are fully developed turbulent flows, and hence Reynold's number is neglected.

Eq. (1) contains the necessary variables to describe the pressure-flow scour under the assumptions of constant fluid density and viscosity and uniform sediment properties. Another dimensionless group, V_b/V_c , obtained by dividing the third group by the second group in Eq. (1), has been defined as the flow intensity ratio by Lyn (2008) ("Discussion" section) and has been used by Arneson (1997)

and Lyn (2008) as an independent variable. The different experimental variables and parameters of the studies on pressure-flow scour and are summed up in Table S5. The table also lists the number of experiments, range of independent and dependent variables, and duration of the experiments.

Equilibrium Time and Temporal Evolution of Pressure-Flow Scour

The process of scouring is time-dependent. The temporal evolution of scouring is important to study because the duration of flood at a bridge site causing pressure-flow scour is usually less than the time to attain equilibrium (Carnacina et al. 2019b; Zhai 2010). This argument remains valid for upstream clear-water cases but practically, during a flood, the upstream conditions will be live-bed, and a much shorter time to equilibrium is expected. This has been well demonstrated [e.g., in Fig. 1 of Melville and Chiew (1999)] for pier scour. But for pressure-flow scour, no study on the variation of time to equilibrium t_e with V_a/V_c has been reported. The Supplemental Materials offer further discussion of the experimental studies on temporal evolution of pressure-flow scour, equilibrium time, and the possible factors that could determine the time to equilibrium.

Perhaps because clear-water experiments are easier to handle in laboratory and provide a conservative estimate, most of the studies on pressure-flow scouring are clear-water studies. This comes with a constraint that the clear-water experiments have to be performed for much longer durations to attain equilibrium (Guo et al. 2009; Hahn and Lyn 2010; Kumcu 2016; Zhai 2010) (Table S5). Earlier studies like those of Abed (1992) and Umbrell et al. (1998) have reported much shorter time durations. Umbrell et al. (1998) conducted experiments for 3.5 h and then extrapolated results to get the equilibrium pressure-flow scour depth. Hahn and Lyn (2010) reported that even after 48 h of experimentation, significant scouring was taking place. Shan et al. (2012) and Kocyigit and Karakurt (2019) arbitrarily defined the equilibrium as the condition when changes in scouring at a reference point are less than 1 mm continuously for 3 and 1 h, respectively, which yielded a time of 32 and 8 h, respectively. A discussion on the effect of duration of the experiments on the model development is presented in the Supplemental Materials.

A proper timescale, T , is needed to nondimensionalize the time variable. The nondimensionalization of time is required to scale up the time to equilibrium from the laboratory to the field conditions. The proposed formulations for the timescale and temporal evolution models are listed in Table 2. The different timescales refer to different properties of the pressure-flow scour phenomenon. Nevertheless, a proper timescale would be such that the dimensionless time to attain the equilibrium pressure-flow scour is independent of the scale of the experiment. The timescale in pressure-flow scouring due to vertical contraction has been developed in two ways. The first method is motivated by the scaling arguments applicable to the horseshoe vortex formed in pier scouring (Melville and Chiew 1999). Melville (2008) defined the timescale for the pier scouring in open-channel condition as $T_p = D/V_a$. A similar timescale for pressure-flow scour due to vertical contraction was used by Zhai (2010) and Carnacina et al. (2019b) in their studies, replacing the pier diameter, D , with a suitable length scale (Table 2).

The second method for timescaling, proposed by Hahn and Lyn (2010), is based on the balance of sediment volume. They argued that the amount of the contraction, H , is a suitable length scale for a scour hole, and the total sediment transported out of the contraction per unit width per unit time, q_s , can be given by, $q_s \sim \theta_a^m \sqrt{g(S_g - 1)D_{50}^3}$, where θ_a is the Shields parameter,

Table 2. Statistical models of the temporal evolution of pressure-flow scour due to vertical contraction

References	Timescale formulation	Magnitude of timescale ^a	Temporal evolution model	Model applicability
Hahn and Lyn (2010)	$T_H \sim \frac{H^2}{\theta_a^2 \sqrt{g(S-1)D_{50}^3}}$	9.7 h	—	Pressure-flow scour
Zhai (2010) ^b	$T_Z \sim \frac{CH_b}{V_a}$	43.3 h	$\frac{y_p}{y_{pe}} = k \left[\ln \frac{tV_a}{CH_b} - 0.33 \left(\frac{tV_a}{CH_b} \right)^3 + \frac{1}{3} \right] + 1$	Pressure-flow scour
Kumcu (2016)	$T_K \sim \frac{H^2}{\theta_a^{2.5} \sqrt{g(S-1)D_{50}^3}}$	78.5 h	$\frac{y_p}{H} = 1.28 \left(\frac{t}{T_K} \right)^{0.05}$	Pressure-flow scour
Carnacina et al. (2019b)	$T_C \sim \frac{DH + BH}{H_a V_a}$	0.41 s ^c	$\frac{y_{pp}}{D} = \xi \ln \left(\frac{0.1t}{T_C} \right)$	Pier scour in pressure-flow conditions

Note: For the definition of ξ , the readers are referred to Eq. (9) in Carnacina et al. (2019b).

^aThe column is meant to give an idea about the magnitude of the timescale. The values might change with the change in experimental variables. The values in this column correspond to the following experimental conditions: $H_a = 0.105$ m, $H_b = 0.083$ m, $V_a = 0.23$ m/s, and $D_{50} = 0.6$ mm.

^b k is a constant defined as $\partial y_p / \partial t = k y_{pe} / t$. Zhai (2010) reported that k depends on the median sediment size, D_{50} ; k was reported to increase with D_{50} . For equilibrium, C was reported to be approximately equal to 4.32×10^5 .

^cThis timescale has been developed for pier scouring in pressure-flow conditions. This value corresponds to the condition when pier diameter, D , is zero.

i.e., $\theta_a = V_*^2 / [g(S_g - 1)D_{50}]$, and V_* is the shear velocity at the upstream of the contraction. Different researchers have suggested different values of the exponent m ranging from 1.5 to 3 (Dey 2011; Madsen 1991; Meyer-Peter and Müller 1948; van Rijn 1984).

The final equation for the timescale defined by Hahn and Lyn (2010) is listed in Table 2. They further reported that the development of a scour hole follows a logarithmic growth but did not provide any expression for it. Zhai (2010) developed a semiempirical model (Table 2) of the time evolution of the scour process based on the sediment continuity equation, i.e., Exner's equation (Coleman and Nikora 2009) and the Meyer-Peter and Müller bed-load transport model (Meyer-Peter and Müller 1948).

An estimate of the time-scale using the aforementioned models for an experimental condition described by Hahn and Lyn (2010) is presented in Table 2. The timescale varies from 0.41 s by Carnacina et al. (2019b) to 78.5 h by Kumcu (2016). The timescale of Hahn and Lyn (2010) yields a physically possible timescale, but the choice of m and length scale for scour hole can lead to significantly different results. For example, Hahn and Lyn (2010) and Kumcu (2016) selected m as 2 and 2.5, resulting in timescales of 9.7 and 78.5 h, respectively. This sensitivity of timescales to exponent m estimated by different models for the example experimental condition suggests the need for more reliable and consistent formulation for the timescale. Besides, as can be seen in the Supplemental Materials, the timescales developed by Hahn and Lyn (2010) and Kumcu (2016) predicted that the equilibrium time, t_e increases with increase in sediment size, D_{50} , which contradicts the experimental results by Zhai (2010) and the Laursen extrapolation model used by Umbrell et al. (1998).

Magnitude of Pressure-Flow Scour

All the models to estimate y_{pe} are listed in Table 3. The mathematical formulations of these models and their identifiers are also given in Table 3. A brief introduction to these models is presented in this subsection. Arneson (1997) conducted unsubmerged clear-water and live-bed experiments to develop statistical model for y_{pe} . The model was also reported by Arneson and Abt (1998). The model was included in the Hydraulic Engineering Circular No. 18 (HEC-18, hereafter referred to as HEC-18b) (Richardson and Davis 2001). Umbrell et al. (1998) studied the pressure-flow scour in clear-water conditions. For some of the experiments, the discharge

was passed only under the bridge and for rest of the experiments, the bridge was completely submerged [Fig. 1(b)].

A model using the one-dimensional continuity equation and incipient velocity, V_c was developed to estimate y_{pe} that has two empirical constants to fit the experimental data. Verma et al. (2004) reported that at critical velocity, V_c , y_{pe} is equal to H . Lyn (2008) reexamined the HEC-18b equation (Richardson and Davis 2001). A spurious correlation was reported by Lyn (2008) in the original analysis (HEC-18b) due to the presence of y_{pe}/H_a on both sides of the equation (a detailed discussion is presented in the "Statistical Testing of Models" section). Consequently, Lyn (2008) fitted a power-law model on the Arneson (1997) data corresponding to the constraints $V_b/V_c < 2$ and $V_a/V_c < 0.9$ and having no pier effect. Lyn (2008) also proposed a design equation for the estimation of y_{pe} .

Guo et al. (2009) conducted both experimental and theoretical studies on unsubmerged pressure-flow scour at $V_a/V_c < 1$. It was reported that the design pressure-flow scour, y_{pe} , decreases with increase in D_{50} . However, the effect of D_{50} is not direct, but indirect through V_c , as seen in Guo et al.'s (2009) model (G2008D in Table 3). In the theoretical analysis, Guo et al. (2009) argued that the bridge scour can be divided into three cases, i.e., downstream unsubmerged, partially submerged, and totally submerged. For downstream unsubmerged flows, the maximum bridge scour depth is an open-channel flow problem where the conventional methods in terms of V_c or τ_{cr} can be applied. For partially and totally submerged flows, a theoretical equation (G2008P) for predicting the scour depth in terms of inundation Froude number was proposed. Hahn and Lyn (2010) studied the unsubmerged clear-water pressure-flow scour. They reported that the observed y_{pe} was substantially more than that predicted by A1997P and U1998P.

Shan et al. (2012) developed an analytical model, S2012P, for y_{pe} when $V_a/V_c < 1$ (Table 3). The model is a refinement over the U1998P in that it is also based on the continuity equation of flow and equilibrium scour depth at critical velocity. The two refinements in S2012P over U1998P are (1) S2012P accounts for flow separation from the leading edge of the bridge deck that creates a flow separation zone (ineffective flow area), t_s , forcing the flow in the bridge opening to contract and accelerate [Fig. 1(b)], and (2) S2012P uses the critical velocity based on the flow under the bridge. Particle image velocimetry (PIV) and computational fluid dynamics (CFD) data, along with the data from Arneson (1997),

Table 3. Prediction and design models for the equilibrium scour depth due to vertical contraction

Model	Identifier	Model formulation	Remarks
Arneson and Abt (1998) ^a	A1997P	$\frac{y_{pe}}{H_a} = -0.93 + 0.23 \frac{H_a}{H_b} + 0.82 \frac{y_{pe} + H_b}{H_a} + 0.03 \frac{V_b}{V_c}$	—
Umbrell et al. (1998) ^a	U1998P	$\frac{y_{pe} + H_b}{H_a} = 1.102 \left[\frac{V_a}{V_c} \left(1 - \frac{w}{H_a} \right) \right]^{0.603}$	—
Lyn (2008) ^a	L2008P	$\frac{y_e}{H_a} = \min \left[0.105 \left(\frac{V_b}{V_c} \right)^{2.95}, 0.5 \right]$	—
Guo et al. (2009) ^{a, b}	G2009P G2009D ^c	$\frac{H_b + y_{pe}}{H_b + a} = \sqrt{\frac{1 + \left(\frac{1.368}{F_i^{2.409}} \right)}{1 + \left(\frac{1.866}{F_i^2} \right)}}, F_i = \frac{V_{ce}}{\sqrt{g(H_a - H_b)}}, a = H_a - (H_b + w)$	$V_{ce} = V_c \left(\frac{H_a - w}{H_a} \right)^{0.85}$
Shan et al. (2012) ^a	S2012P	$y_{pe} = \left(\frac{V_a(H_a - w)}{K_U D_{50}^{1/3}} \right)^{\frac{6}{5}} + \left[0.2 \left(\frac{H_b H}{H_a^2} \right)^{0.2} \left(1 - \frac{w}{H} \right)^{-0.1} - 1 \right] H_b$	$K_U = 6.17 \text{ m}^{1/2}/\text{s}$
Kocyigit and Karakurt (2019) ^a	KK2019P	$\frac{y_{pe}}{H_b} = -0.962 - 0.187 \frac{H_a}{H_b} + 0.443 F_b^* + 0.672 \frac{G}{H_b}, F_b^* = \frac{H_b}{\sqrt{g(S_g - 1)D_{50}}}$	Unsubmerged pressure flow
		$\frac{y_{pe}}{H_b} = -1.002 - 0.190 \frac{H_a}{H_b} + 0.439 F_b^* + 0.925 \frac{G}{H_b}$	Submerged pressure flow
Lyn (2008) ^b	L2008D	$\frac{y_{pe}}{H_a} = \min \left[0.21 \left(\frac{V_b}{V_c} \right)^{2.95}, 0.6 \right]$	—
Shan et al. (2012) ^b	S2012D	$y_{pe} = \left(\frac{V_a(H_a - w)}{K_U D_{50}^{1/3}} \right)^{\frac{6}{5}} + \left[0.5 \left(\frac{H_b H}{H_a^2} \right)^{0.2} \left(1 - \frac{w}{H} \right)^{-0.1} - 1 \right] H_b$	$K_U = 6.17 \text{ m}^{1/2}/\text{s}$
Umbrell et al. (1998) ^b	U1998D	$\frac{y_{pe} + H_b}{H_a} = 1.102 \left[\frac{V_a}{V_c} \left(1 - \frac{w}{H_a} \right) \right]^{0.603} + 0.06$	—
Melville (2014) ^b	M2014D	$\frac{y_{pe}}{H_a} = \begin{cases} 0.75 \left(\frac{V_a}{V_c} - 0.4 \right), & 0.4 < \frac{V_a}{V_c} \leq 1 \\ 0.45, & 1 < \frac{V_a}{V_c} \leq 2.5 \end{cases}$	—
Kumcu (2016) ^b	K2016D	$\frac{y_{pe} + H_b}{H_a} = \begin{cases} 0.65 + 0.5 \frac{V_b}{V_c}, & 0.5 \leq \frac{V_b}{V_c} < 1 \\ 1.025 + 0.125 \frac{V_b}{V_c}, & 1 \leq \frac{V_b}{V_c} \leq 1.8 \end{cases}$	—

Note: For all models that use V_c explicitly, $V_c = C_{vc} \sqrt{g(S_g - 1)D_{50}(H_a/D_{50})^{\frac{1}{6}}}$, $C_{vc} = 1.52$, except for Umbrell et al. (1998) where, $C_v = 1.58$.

^aPrediction model.

^bDesign model.

^cThe model developed by Guo et al. (2009) is a prediction model when approach flow velocity, V_a is used and becomes a design model when V_c is used in the model.

Umbrell et al. (1998), and their own experiments, were used to estimate t_s as a function of flow geometry. Shan et al. (2012) also proposed their own design model, S2012D, and modified the U1998P model to make it a design model, U1998D. S2012D has replaced the A1997P in the latest HEC-18 manual (Arneson et al. 2012). HEC-18 recommends S2012D for clear-water conditions (for which the model was developed) as well as for live-bed conditions with a slight modification.

Melville (2014) conducted submerged and unsubmerged live-bed experiments for pressure-flow scour and developed design relations for y_{pe} . Kumcu (2016) carried out the study of unsubmerged clear-water pressure-flow scour under both steady and unsteady conditions and developed design relations. It was reported that y_{pe} increases with V_b/V_c ; y_{pe} for almost all unsteady tests with peak discharge, Q_p , were less than y_{pe} for the steady-flow conditions with discharge Q_p . Kocyigit and Karakurt (2019) studied the submerged and unsubmerged clear-water pressure-flow scour, and models were developed for unsubmerged and submerged pressure-flow scour, respectively. It was reported that densimetric Froude

number, $F_b^* = V_b / \sqrt{g(S_g - 1)D_{50}}$, is the most significant for predicting y_{pe} . F_b^* can be reduced to V_b/V_c (Carnacina et al. 2019b), which is the only dimensionless group used in L2008P. Hence, the findings reported by Kocyigit and Karakurt (2019) are consistent with the results and model reported by Lyn (2008).

Geometry and Location of Pressure-Flow Scour Hole

Guo et al. (2009) reported that the profile of the pressure-flow scour hole is two-dimensional. One of most significant findings of this study was the development of similarity equations for the geometry of the scour hole [Eq. (2)] (Fig. 2); x is measured from the upstream end of the contraction, and y is measured vertically upward from initially flat bed

$$\frac{y}{y_{pe}} = \begin{cases} -e^{-\left(\frac{x}{L} - 0.85\right)^{2.5}}, & \frac{x}{L} \leq 0.85 \\ -1.055e^{-\left[\frac{x}{L} - 0.85\right]^{1.8}} + 0.055, & \frac{x}{L} > 0.85 \end{cases} \quad (2)$$

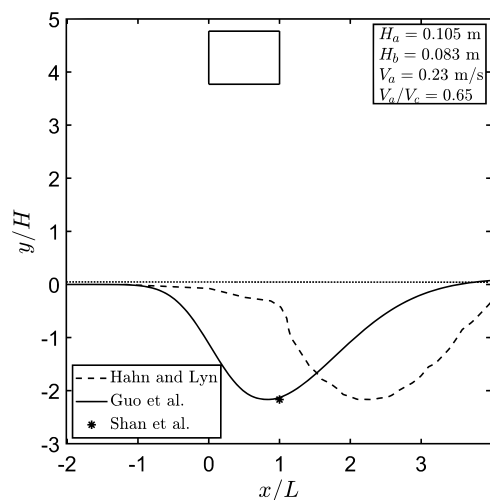


Fig. 2. Comparison of scour profile observed by Hahn and Lyn (2010) with the estimation by the Guo et al. (2009) model [Eq. (2)] and maximum scour location predicted by Shan et al. (2012).

Hahn and Lyn (2010) observed that the scour hole developed after the downstream end of the bridge and was considered anomalous (Fig. 2). Shan et al. (2012) reported that under clear-water conditions, the maximum scour occurred at the downstream end of the bridge deck.

Fig. 2 shows the experimental profile reported by Hahn and Lyn (2010) and the predicted profile by Guo et al. (2009) for the same experimental conditions. The location of maximum scour reported by Shan et al. (2012) is also shown. It is clear that there is a significant difference in the predicted location of maximum scour by Guo et al. (2009) and Shan et al. (2012) from that of observed location of maximum scour by Hahn and Lyn (2010). Interestingly, Hahn and Lyn (2010) reported it as an anomaly because the observed location is outside of the contraction. This anomaly questions the one-dimensional flow assumption by Umbrell et al. (1998) and motivates future researchers to investigate the nature of flow in pressure-flow scouring.

Velocity Field

Hahn (2005) studied the velocity field in pressure-flow scour due to vertical contraction using laser Doppler velocimetry (LDV). The vertical velocity profile along the centerline was studied. At the location of maximum scour, i.e., downstream of the bridge, maximum velocity was observed to be at the center of the vertical profile. Hahn (2005) also reported that the scouring occurred even though average velocity, V_a or V_b , did not exceed the critical velocity V_c at any section.

Carnacina et al. (2019a) studied the characteristics of flow structure around cylindrical bridge piers in pressure-flow conditions using acoustic Doppler velocimetry (ADV). It was observed that the maximum average velocity components are accelerated due to vertical contraction, and this acceleration is further increased by the development of a boundary layer under the bridge deck. Such an observation was also reported by Shan et al. (2012). Yoon et al. (2019) studied the time-averaged turbulent velocity field near the bridge abutments in pressure-flow conditions due to vertical contraction using an ADV. It was reported that pressure flow imparts additional flow acceleration close to the abutment, resulting in the maximum scour closer to it.

Statistical Testing of Models

Table S6 summarizes the existing prediction and design models and the data sets used in their development. The existing models can be categorized on different lines. Prediction models are developed by fitting a best-fit model to a given experimental data set (A1997P, U1998P, L2008P, G2009P, and KK2019P), and design models envelop most of the experimental data (L2008D, S2012D, M2014D, and K2016D). The prediction models can be purely empirical in nature where the significant variables are determined by examining the experimental data, as has been done in A1997P, L2008P, and KK2019P. On the other hand, U1998P, S2012P, and G2009P are driven by a theoretical basis but rely on the experimental data to train some parameters of the model. All the design models except S2012D and G2009D have been developed by fitting an empirical envelope to the available data.

The models can further be classified based on the choice of the dependent variables. All the empirical models except KK2019P and K2016D have chosen H_a as the appropriate length scale to nondimensionalize y_{pe} , perhaps because H_a is a readily available variable. U1998P has used $(y_{pe} + H_b)/H_a$, the ratio of total clearance under the bridge after scour to the upstream flow depth, as the dependent variable based on continuity equation. Among the independent variables, A1997P, L2008P, L2008D, and K2015D have used V_b/V_c as the velocity ratio, whereas U1998P and M2014D have used V_a/V_c . KK2019P has used an additional group, F_b^* , which, as previously mentioned, can be reduced to V_b/V_c .

Performance of the Models

Considering the one-dimensional continuity equation and sediment transport models, it is fair to assume that y_{pe}/H_a should increase with the increase in V_a/V_c and decrease with increase in H_b/H_a for clear-water conditions. In fact, except for A1997P, all other models predict the same, as is shown in Figs. 3(a and c), respectively. A1997P predicts a considerable amount of deposition in the range of H_b/H_a from 0.5 to 0.7 for $V_a/V_c = 1$, and y_{pe} increases as H_b/H_a increases, which is counterintuitive. This was later reported to be an outcome of spurious correlation in Arneson's model, A1997P (Lyn 2008).

In the asymptotic conditions, when H_b/H_a tends to unity, especially when V_a/V_c is far less than unity, U1998P and S2012P equations predict deposition, which is not possible for upstream clear-water conditions [Fig. 4(c)]. This limitation can be overcome by redefining the model and forcing no-scour and no-deposition conditions when H_b/H_a tends to unity. It is also observed that L2008P predicts positive scour values for all possible values of V_b/V_c ; however, y_{pe}/H_a is negligible (<0.1) for $V_b/V_c < 1$. Kumcu (2016) also reported that there was no pressure-flow scouring due to vertical contraction in the experiment having a value of $V_b/V_c = 0.86$.

It can be observed in Figs. 3(b and d) that the design scour depth from M2014D is independent of H_b/H_a and depends only on V_a/V_c . Melville (2014) did recognize the dependence of y_{pe}/H_a on H_b/H_a but chose a simple conservative envelope curve for the design model that did not depend on H_b/H_a . Melville (2014) suggested that additional research is needed to add a factor for the effects of H_b/H_a on y_{pe}/H_a . The design model K2016D is more sensitive to the contraction ratio, H_b/H_a , than V_a/V_c , and L2008D shows moderate dependence on both ratios H_b/H_a and V_a/V_c . Besides, in the asymptotic conditions when $V_a/V_c = 1$ and H_b/H_a tends to unity, K2016D and L2008D collapse and predict y_{pe} of magnitudes 15% and 21% of the H_a , respectively.

Due to scarcity of the available data, the data corresponding to the studies by Arneson (1997), Umbrell et al. (1998), Guo et al.

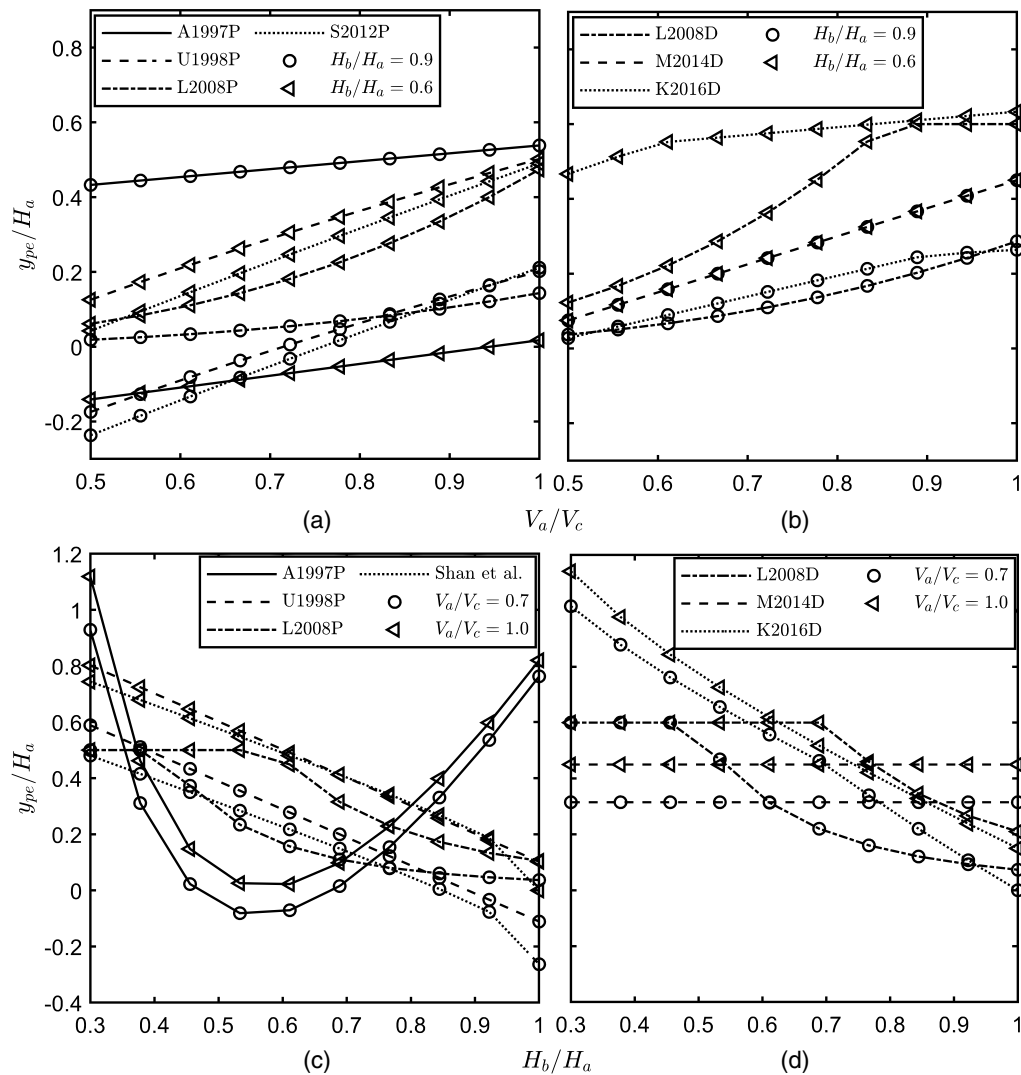


Fig. 3. Plot of the prediction and design models: (a) variation of best-fit models with V_a/V_c for constant values of H_b/H_a ; (b) variation of design models with V_a/V_c for constant values of H_b/H_a ; (c) variation of best-fit models with H_b/H_a for constant values of V_a/V_c ; and (d) variation of design models with H_b/H_a for constant values of V_a/V_c .

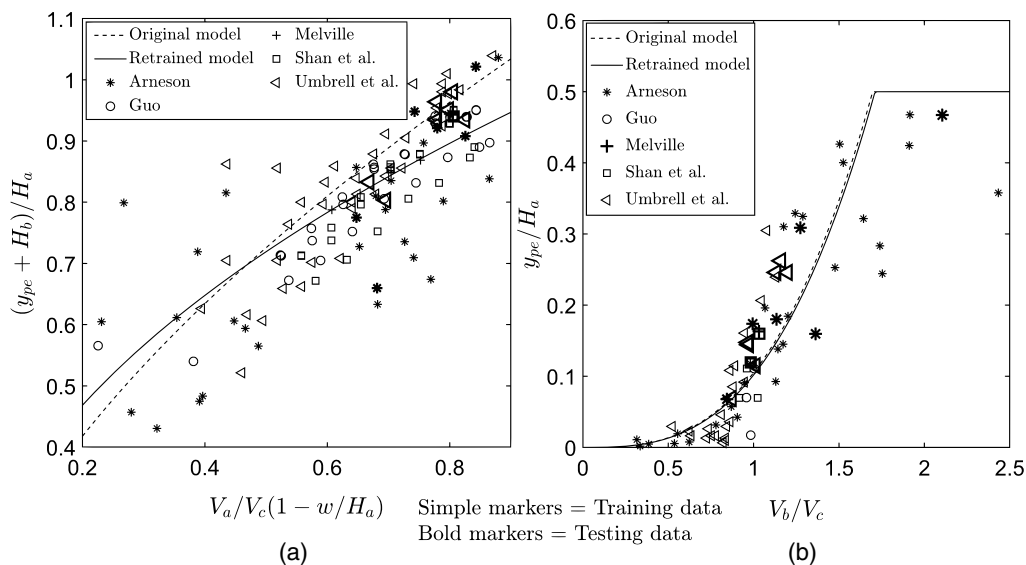


Fig. 4. Retraining of the models: (a) Umbrell et al. (1998); and (b) Lyn (2008).

(2009), Shan et al. (2012), and Melville (2014) were combined to get a data set having 201 experimental points. A common data set of 51 points was filtered to test the existing prediction models. The data set contains 22, 23, 4, 1, and 1 data points from Arneson (1997), Umbrell et al. (1998), Shan et al. (2012), Guo et al. (2009), and Melville (2014), respectively. The selected data set satisfied the constraints of all the prediction models (no pier, $w/H_a = 0$, $V_a/V_c < 0.95$, and $V_b/V_c < 2$). Besides, only those points for which $y_{pe}/H_a > 0.1$ are used for testing (Lyn 2008). Similarly, a data set of 91 points [26, 51, 3, 6, and 3 data points from Arneson (1997), Umbrell et al. (1998), Guo et al. (2009), Shan et al. (2012), and Melville (2014), respectively] was filtered out to test the design models, which satisfied the constraints of all the design models (no pier, $w/H_a = 0$, $V_a/V_c > 0.4$, $V_b/V_c > 0.5$, and $V_b/V_c < 1.8$) and for which $y_{pe}/H_a > 0.1$. For testing M2014D, $H_b/H_a > 0.5$ was also enforced in addition to the preceding constraints.

The statistical metrics of the prediction and design models on the testing data sets are presented in Table 4. The performance is reported in terms of root mean squared error (RMSE) and mean and standard deviation of discrepancy ratio, P_r , defined as $(y_{pe}/H_a)_{\text{predicted}}/(y_{pe}/H_a)_{\text{observed}}$. P_r is quite sensitive to smaller values of the observed data and should be interpreted carefully. Further, large deviations from $P_r = 1$ for small values of y_{pe}/H_a may be of little practical significance, but even moderate deviations from $P_r = 1$ for large values of y_{pe}/H_a may have serious practical consequences. In this study, the excessive sensitivity of P_r is to some extent mitigated by filtering out the points having very small values of $(y_{pe}/H_a)_{\text{observed}}$.

Among the prediction models, the A1997P model had the worst performance during testing. It predicts a significant number of depositions. G2009P and KK2019P also did not perform well,

as is evident from their RMSE and other statistics in Table 4. G2009P is a physically based model developed using the continuity and energy equations for water flow. The model behaves as a prediction model, G2009P, or a design model, G2009D, when V_a or V_c is used in the model, respectively. The model parameters in G2009P have been trained on the scour data with V_a very close to incipient conditions (V_a/V_c between 0.8 and 0.9). The model, as expected, performs relatively well for the test data with V_a/V_c between 0.8 and 0.9 (RMSE = 0.079, mean of $P_r = 1.17$, and standard deviation of $P_r = 0.35$), whereas for the data with $V_a/V_c < 0.8$, the model largely overpredicts the data (RMSE = 0.360, mean of $P_r = 2.42$, and standard deviation of $P_r = 1.43$).

Overall, the model G2009P tends to overpredict the data with mean of $P_r = 1.67$. KK2019P is an empirical model, and one dimensionless parameter, G/H_b , has been used as an independent variable. The actual girder depth, G , is not reported in some of the studies (Arneson 1997; Melville 2014; Umbrell et al. 1998). To address this issue, a typical two-lane bridge is assumed to have a girder of depth 1.22 m [Design example 8.1 in Richardson and Davis (2001)], and scaling it down to the laboratory gave the respective girder depths (model scales for Arneson and Umbrell are 1:8 and 1:15, respectively). Nevertheless, Kocyigit and Karakurt (2019) reported that F_b^* is the most influential parameter in their model.

In the testing, it appears that KK2019P largely overpredicts the data with mean of $P_r = 3.41$. The other three prediction models show a relatively good performance, with S2012P having the lowest RMSE, mean of P_r close to unity, and more fraction of data with $0.67 < P_r < 1.5$. This is obvious because the testing data set is dominated by the Arneson (1997) and Umbrell et al. (1998) data, the same data that were originally used to train the model, S2012P. It can be seen in Table 4 that 49 out of 51 testing data points were also

Table 4. Summary of the statistical performance of prediction and design models

Model	Empirical basis of data set		Performance statistics ^a on testing							Percentage of predictions with P_r greater than		
	Training	Testing	RMSE on testing data	Percentage of depositions	Percentage of testing data with $0.67 \leq P_r \leq 1.5^b$	Mean of P_r	Standard deviation of P_r	Percentage of underpredictions		1	2	3
A1997P	AA-116	AA-22, U-23, M-1, S-4, G-1	0.24	15.7	15.7	1.11	1.34	—		—	—	—
U1998P	U-81	- do -	0.08	0	74.5	1.14	0.43	—		—	—	—
L2008P	AA-33	- do -	0.09	0	62.8	0.89	0.36	—		—	—	—
G2009P	G-28	- do -	0.26	2.0	43.1	1.67	1.28	—		—	—	—
S2012P	AA-30, U-53, S-26	- do -	0.07	0	76.5	0.92	0.37	—		—	—	—
KK2019P	KK-102	AA-22, U-23, S-4, G-1	0.53	0	2.0	3.41	1.44	—		—	—	—
L2008D	—	AA-26, U-51, S-6, G-3, M-3	—	—	—	—	—	1.1		98.9	42.7	22.5
G2009D	—	- do -	—	—	—	—	—	53.9		46.1	25.8	14.6
S2012D	—	- do -	—	—	—	—	—	4.5		95.5	46.1	23.6
U1998D	—	- do -	—	—	—	—	—	7.9		92.1	23.6	14.6
M2014D	—	AA-20, U-51, S-6, G-3, M-3	—	—	—	—	—	6.0		94.0	54.2	36.1
K2016D	—	AA-26, U-51, S-6, G-3, M-3	—	—	—	—	—	0		100	57.3	36.0

Note: AA-116 means that 116 datapoints from Arneson (1997) have been used in the training of the model and so on. AA = Arneson and Abt; G = Guo et al.; M = Melville; S = Shan et al.; U = Umbrell et al.; KK = Kocyigit and Karakut; and - do - = same as above.

^aAll the statistics are calculated on y_{pe}/H_a . The prediction models are tested for the percentage of negative predictions, the fraction of the data with P_r in the range of 0.67 and 1.5, the bias of P_r and the SD of P_r (Lyn 2008). The design models are tested for their ability to overpredict the data and the range of overpredictions (Melville and Lim 2014).

^b P_r is defined as $(y_{pe}/H_a)_{\text{predicted}}/(y_{pe}/H_a)_{\text{observed}}$.

used in the training of the model. On the other hand, U1998P and L2008P perform well on the testing data set, although approximately 45% of the testing data was used to train the original model.

In Fig. S5, it can be observed that the better performance of U1998P and L2008P is partly due to the presence of testing data points from their respective training data sets. U1998P has lower RMSE and higher fraction of data having P_r within 2/3 and 3/2 compared with L2008P, but the standard deviation of P_r is better for L2008P compared with U1998P.

All the design models except G2009D provide conservative estimates, as reflected in the low percentage of test data that are underpredicted. Both L2008D and S2012D underpredict less than 5% of the test data; however, L2008D is better than S2012D in terms of percentage of test points overpredicted by more than 100%, i.e., $P_r > 2$, and 200%, i.e., $P_r > 3$. U1998D also exhibits good performance with less than 8% of underpredictions. M2014D has relatively more underpredictions ($P_r < 1$) and overpredictions by more than 200% ($P_r > 3$) compared with L2008D and S2012D. Compared with other design models, K2016D has the highest number of overpredictions by more than 200% ($P_r > 3$). G2009D has the worst metrics, with about 54% of underpredictions.

Retraining of the Prediction Models

The Arneson (1997), Umbrell et al. (1998), Lyn (2008), Guo et al. (2009), Shan et al. (2012), and Kocyigit and Karakurt (2019) prediction models were retrained and then tested on a common independent data set. They will be hereafter referred to as A1997R, U1998R, L2008R, G2009R, S2012R, and KK2019R, respectively. These models are selected because they are among the most popular models that lend themselves to retraining. Shan et al.'s (2012)

model consists of two sets of parameters, one obtained by fitting the model to the observed scour data and the other obtained based on PIV and CFD data for the stagnation zone thickness, t_s . In retraining, we have retained the set of parameters for t_s based on PIV and CFD data and retrained the remaining parameters.

The originally trained equation is given as Eq. (3). The exponent of last term of right hand side is adjusted based on PIV and CFD Data to -0.1 . Thus, in the retraining, the constant 0.2 and exponent of $(1 - w/H_a)$ are retained, and the exponents of H_b/H_a and H/H_a are retrained. The retrained model for t_s/H is given in Eq. (4). The final model S2012R so developed using Eq. (4) is presented in Table 5

$$\frac{t_s}{H} = 0.2 \left(\frac{H_b}{H_a} \right)^{1.2} \left(\frac{H}{H_a} \right)^{-0.8} \left(1 - \frac{w}{H} \right)^{-0.1} \quad (3)$$

$$\frac{t_s}{H} = 0.2 \left(\frac{H_b}{H_a} \right)^{0.94} \left(\frac{H}{H_a} \right)^{-0.88} \left(1 - \frac{w}{H} \right)^{-0.1} \quad (4)$$

The retraining is aimed to remove any bias in statistical comparison of the models arising due to the data sets used in developing the original models. Due to scarcity of the available data, the data corresponding to the studies by Arneson (1997), Umbrell et al. (1998), Guo et al. (2009), Shan et al. (2012), and Melville (2014) are combined. Data having the pier effect and negative scour values are removed to get a total of 201 experimental points. A subset from that data having a total number of 50 data points and satisfying the constraints ($V_a/V_c < 0.9$, $w/H_a = 0$, $y_{pe}/H_a > 0.05$) is filtered out, and 32% (16 data points) of this data set is randomly separated for the testing purposes. Due to the limited number of testing data

Table 5. Summary of the statistical performance of retrained prediction models, the data sets used in the training and testing of these models, and comparison with original models

	A1997R ^a	U1998R ^b	L2008R ^c	G2009R ^d	S2012R ^e	KK2019R ^f
Number of data points used in retraining of the model	94 (AA–39, U–44, S–4, M–4, G–3)	123 (AA–28, U–39, S–24, M–3, G–29)	56 (AA–28, U–21, S–4, G–3)	147 (AA–32, G–28, M–3, S–24, U–60)	119 (AA–18, U–46, S–24, M–3, G–28)	90 (AA–39, G–3, S–4, U–44)
Method for retraining	CLR	LTLR	LTLR	DNLR	DNLR	CLR
R^2 of the retrained model	0.87	0.55	0.74	0.31	0.84	0.83
Number of data points used in testing of the retrained and original model ^g	16 (AA–6, U–7, M–1, S–2)					15 (AA–6, U–7, S–2)
RMSE on testing data	(R) 0.222	0.069	0.063	0.088	0.046	0.070
	(O) 0.233	0.067	0.061	0.180	0.052	0.517
Percentage of deposition predictions	(R) 43.8	12.5	0	0	0	0
	(O) 12.5	0	0	0	6.3	0
Percentage of testing data with $0.67 \leq P_r \leq 1.5^h$	(R) 18.8	62.5	62.5	87.5	81.2	73.3
	(O) 12.5	81.3	62.5	56.3	68.8	0
Mean of P_r	(R) 0.65	0.77	0.81	1.33	0.97	0.94
	(O) 1.58	1.18	0.83	1.61	0.81	4.14
Standard deviation of P_r	(R) 1.26	0.54	0.26	0.50	0.36	0.32
	(O) 2.19	0.40	0.26	0.90	0.42	1.88

Note: All the statistics are calculated on y_{pe}/H_a . AA–39 means that 39 data-points from Arneson (1997) have been used and so on. (R) = retrained model; (O) = original model; AA = Arneson and Abt; U = Umbrell et al.; S = Shan et al.; M = Melville; G = Guo et al.; HL = Hahn and Lyn; CLR = classical linear regression; LTLR = log-transformed linear regression (Lyn 2008); and DNLR = direct nonlinear regression (Seber and Wild 2003).

^a $y_{pe}/H_a = -0.96 + 0.26H_a/H_b + 0.78[(H_b + y_{pe})/H_a] + 0.03V_b/V_c$.

^b $(y_{pe} + H_b)/H_a = 1.00\{(V_a/V_c)[1 - (w/H_a)]\}^{0.47}$.

^c $y_{pe}/H_a = \min[0.101(V_b/V_c)^{2.96}, 0.5]$.

^d $(H_b + y_{pe})/(H_b + a) = \sqrt{[1 + (0.015/F_i^{0.534})]/[1 + (0.010/F_i^2)]}$.

^e $y_{pe} = \{[V_a(H_a - w)]/6.17D_{50}^{1/3}\}^{6/7} + [0.2(H/H_b)(H_b/H_a)^{0.94}(H/H_a)^{-0.88}[1 - (w/H^{-0.1} - 1)]H_b$.

^f $y_{pe}/H_b = -0.52 + 0.49(H_a/H_b) + 0.07F_b^* - 0.60(G/H_b)$.

^gRetrained as well as the original models are tested on a common data set.

^h P_r is defined as $(y_{pe}/H_a)_{\text{predicted}}/(y_{pe}/H_a)_{\text{observed}}$.

points (16 points), it may not be possible to establish a solid statistical basis for differentiating the performance of different models.

The rest of the 185 points constitute the training data set. The constraints in the original models (A1997R: $w/H_a = 0$, U1998P: $V_a/V_c < 1.05$, L2008P: $V_a/V_c < 0.9$, $w/H_a = 0$, and S2012P: $V_a/V_c < 1.0$) are also enforced during the retraining, leading to different numbers of training data points for different models (Table 5). However, all the retrained models were tested on a common filtered data set to avoid any bias. The common test data set satisfied the constraints of all the models (no pier, $w/H_a = 0$, $V_a/V_c < 0.9$, and $V_b/V_c < 2$).

A summary of the retrained models and their statistical performance is presented in Table 5. The original models U1998P and L2008P have R^2 of 0.81 and 0.87, respectively, compared with 0.55 and 0.74 of the retrained models U1998R and L2008R, respectively. U1998P and L2008P have been trained on single data sets, i.e., data sets by Umbrell et al. (1998) and Arneson (1997), respectively. But in retraining, the models were trained on other data sets as well. The significant decrease in the R^2 in U1998R is perhaps due to the variability and inhomogeneities across the data sets due to different durations and scales of experiments in these studies. Nevertheless, L2008P still has a good fit with R^2 0.74.

Because the data from Umbrell et al. (1998) and Arneson (1997) form a major part of the data available and were also used in the training of the S2012R, the good R^2 and statistical performance of the S2012R are expected. Fig. 4 shows the U1998R and L2008R along with the training and the testing data (markers) (A1997R and G2009R are not included in Fig. 4 because of their poor performance and S2012R and KK2019R because they have multiple parameters.). It appears that the $(y_{pe} + H_b)/H_a$ reported in the Umbrell et al. (1998) data is higher than in the rest of the data, especially for higher values of $V_a/V_c(1 - w/H_a)$, resulting in a greater scatter of data, eventually leading to a reduced R^2 of the U1998R model [Fig. 4(a)]. A slight increase in the slope of the L2008R model compared with the original model is also due to aberrations in the Umbrell et al. (1998) data set.

The performance metrics of the original models on the same testing data set are also presented in Table 5. It can be observed that both the A1997P and A1997R are not performing well. The Arneson (1997) model is included only because of its historical interest and its shortcoming are now widely recognized (Lyn 2008). The main motivation of including it is to illustrate that some standard statistical checks might have revealed its shortcomings. The performance metrics of G2009R and KK2019R improve significantly over G2009P and KK2019P because the original models were trained on limited data points (Table 4). U1998P, U1998R, L2008P, L2008R, S2012P, S2012R, and KK2019R are performing well on testing. L2008R is performing almost as well as L2008P on the test data set. S2012R is performing slightly better than the original one.

U1998R, on the other hand, is not performing as well as the original model. A possible reason could be that some of the testing data points were used for training U1998P. Even after retraining, G2009R's performance is not comparable to other models. The performance metrics of the S2012R are the best among all the models; however, because only two of the four model coefficients of S2012R were retrained [Eq. (4)] and the model has already seen most of the test data (15 out of 16 points), it is possible that the performance metrics are biased in its favor. Cross-validation was also performed to increase the robustness of the test results. The results are reported in Tables S3 and S4 and are comparable to those reported in Table 5.

The scarcity of the available data and different constraints imposed by the models were the two major hindrances in retraining the pressure scour prediction models. The Arneson and Abt (1998)

model has been developed for both live-bed and clear-water conditions, but is not performing well, as is evident from the performance statistics (Table 5). The other prediction models (Kocyigit and Karakurt 2019; Lyn 2008; Shan et al. 2012; Umbrell et al. 1998) are performing well on the test data. These models have been developed on clear-water experiments and have not been tested for live-bed conditions. The deviations of the predictions using these models for live-bed conditions is not known; however, they can be conservatively extended to live-bed conditions as well.

On the other hand, the design models (Kumcu 2016; Lyn 2008; Melville 2014) have been developed on and can be applied to both clear-water and live-bed conditions. Overall, the results suggest that more experiments are needed to develop a reliable model for the estimation of pressure-flow scour, y_{pe} , for both clear-water and live-bed conditions. It is worth mentioning that the formulation of S2012R is very complicated against the simple and elegant formulation of L2008R and U1998R. Besides, the retrained coefficients of the models show that L2008R is more robust and the least sensitive to data changes.

Pier Scouring in Pressure-Flow Conditions

Contrary to the pier scour in open-channel conditions, pier scour in pressure-flow conditions has not been studied extensively, and most of its aspects are under question. Abed (1992) studied unsubmerged clear-water pier scouring in pressure-flow conditions [Fig. 1(c)] and reported that the total scour at the pier y_{ppe} was a combined effect of local obstruction at pier and vertical contraction to the flow. It is also reported that y_{ppe} is much larger than y_{ope} (2–10 times) for the same upstream flow conditions, but the differentiated components are not reported.

Pier scour in pressure-flow conditions can be assumed to be composed of two components: (1) due to vertical contraction of the flow $(y_{ppe})^c$; and (2) due to pier obstruction $(y_{ppe})^p$ [Fig. 1(d)]. Most authors have recommended to estimate $(y_{ppe})^p$ using formulations for pier scour in open-channel conditions, i.e., $(y_{ppe})^p = y_{ope}$ and $(y_{ppe})^c$ using models for pressure-flow scour, i.e., $(y_{ppe})^c = y_{pe}$ (Jones et al. 1993; Richardson and Davis 2001; Umbrell et al. 1998). Arneson (1997) suggested to estimate y_{ppe} by adding y_{pe} , calculated using A1997P, and y_{ope} , calculated for the case when flow is at the onset of transition to pressure flow.

Similarly, Umbrell et al. (1998) suggested to use U1998P equation to compute the y_{pe} and add it to y_{ope} computed using HEC-18a equation [Eq. (21) in the HEC-18a manual by Richardson et al. (1993)] to find y_{ppe} . Richardson and Davis (2001) provided a guideline to estimate y_{ppe} by adding y_{pe} calculated using A1997P equation to y_{ope} [Eq. (6.1) in Richardson and Davis (2001)] for the same flow variables. The most recent version of HEC-18 manual (Arneson et al. 2012) provides a recommendation to evaluate y_{pe} using S2012D, but the guidelines to evaluate y_{ppe} are not available in the manual.

However, because of limited experiments, the validity of this assumption is not beyond ambiguity. For example, Arneson (1997) reported that pier scour component $(y_{ppe})^p$ is smaller than the pier scour in open-channel conditions, y_{ope} . Jones et al. (1993) reported $(y_{ppe})^p$ to be almost equal to y_{ope} , and Carnacina et al. (2019b) reported that $(y_{ppe})^p$ is higher than y_{ope} .

Discussion

Statistical models to predict y_{pe} are developed based on laboratory data, but they have not been tested on the field conditions.

Laboratory-trained models tends to overestimate the scour depth for field conditions, as in the case of pier scour (Ettema et al. 1998) due to their inability to maintain a complete similitude. It is still not clear whether such an argument is applicable to pressure-flow scouring. The scaling effects on the y_{pe} have not been reported in any of the studies. Hahn and Lyn (2010), who did experiments on a relatively small flume, reported significantly higher scour depths compared with the predictions by the existing models. The ultimate test for the discrepancy between laboratory and field data requires field observations for pressure-flow conditions, which are presently nonexistent due to difficulties in scour measurement under such conditions.

The equilibrium time is an important issue for laboratory experiments because the prediction and design equations are developed for equilibrium scour under clear-water conditions. The clear-water experiments take significantly higher time to reach equilibrium than the live-bed experiments, and hence the equilibrium time and temporal evolution are more critical for laboratory experiments than field applications. Therefore, prediction of pressure-flow scour depth requires an understanding of the temporal evolution of pressure-flow scour and a timescale to nondimensionalize the time in pressure-flow scour. Two approaches to develop an appropriate timescale have already been discussed previously. The first approach assumes pressure-flow scour to be analogous to open-channel pier scour, for which there is hardly any evidence yet (Carnacina et al. 2019b; Zhai 2010). The second approach assumes applicability of open-channel formulations of sediment transport and inception of sediment motion to pressure-flow conditions (Hahn and Lyn 2010), which is questionable.

The scour hole, resulting from the increased velocity in the contraction, is expected to form within the contraction. An anomaly was reported by Hahn and Lyn (2010), where the scour hole is observed to have formed downstream (outside) of the contraction. This anomaly raises questions about the applicability of inception of motion and scour evaluation models developed for open-channel conditions to pressure-flow conditions. The effect of the channel aspect ratio also remains unexplored and has not been reported in any study. Most of the studies on pressure-flow scouring due to vertical contraction employed wide flumes to avoid the sidewall effects.

The effect of the length of the bridge, L , on the pressure-flow scour depth, y_{pe} , has not been reported in any of the studies. Like the horizontal contractions (Dey and Raikar 2005), long and short vertical contractions may affect scouring. The observation of a stagnation zone by Shan et al. (2012) suggests that if the length of contraction, L , is long enough that the flow inside the bridge stabilizes, then the problem can be treated as long contraction, where equilibrium scour depth, y_{pe} , is independent of the L . Otherwise, the depth of the stagnation zone increases with increase in L (Shan et al. 2012), which could possibly affect the y_{pe} as well as the geometry of the scour hole.

The incipient velocity conditions under pressure-flow conditions have been described using two dimensionless groups, V_a/V_c and V_b/V_c (Arneson and Abt 1998; Lyn 2008; Umbrell et al. 1998). In both the cases, V_c has been calculated based on the upstream section of the vertical contraction. For V_b/V_c , V_b corresponds to the velocity inside the contraction, and hence nondimensionalizing it with V_c calculated for upstream conditions is unconvincing. Employing the open-channel formulation for calculating the V_c in pressure-flow conditions might introduce errors because the velocity distribution in pressure-flow conditions is different from that in open-channel flow conditions (Lyn 2008). Shan et al. (2012) addressed this issue by using the critical velocity, V_c , based on the flow under the bridge.

No significant attempt has been made to study pressure-flow scour numerically. Kara et al. (2015) numerically studied flow dynamics through a submerged bridge opening but for a rigid bed. Although not quantified, it was reported that there is an increase in the bed shear stresses in the contraction. But the inception of the sediment motion and the motion of sediment in pressure-flow conditions are not well understood. Therefore, the use of empirical sediment transport equations, which have been developed for open-channel conditions, in numerical models may not be useful in studying pressure-flow scour due to vertical contraction. However, the two-phase models, where the fluid phase and sediment phase are coupled and solved together and thus do not rely on the empirical sediment motion models, have been successfully used to simulate scouring by submerged wall jets (Lee et al. 2019) and scouring under submarine pipelines (Mathieu et al. 2019). The mechanism of pressure-flow scouring, if not completely identical, is similar to the scouring by submerged wall jets and scouring under submarine pipelines, and therefore the lessons learned from their modeling can help in the modeling of pressure-flow scour.

A complete description of the independent and dependent variables, as discussed in the "Introduction" section, that includes t_e , y_{pe} , y_{ppe} , and the location and geometry of the scour, should be reported in any experimental study, but that has not always been the case in the past. For example, although Arneson and Abt (1998) and Umbrell et al. (1998) provided extensive data, they did not report the location of the scour hole and t_e . Absence of these vital details reduces the usefulness of experimental data for further analysis.

Conclusions

The review of the literature established the importance and underscored the challenges of the pressure-flow scour problem due to vertical contraction. It suggests that the problem cannot be completely addressed using conventional methods of scour estimation, which have been mostly developed for open-channel flow conditions. The literature available provides some hints for addressing the problem. The main conclusions from the study are as follows:

- The present pressure-flow scour data need to be increased by conducting experiments on wider ranges of parameters that are more representative of field conditions. For example, most of the present studies were conducted for upstream clear-water conditions, whereas live-bed conditions are prevalent in field. Further, the existing data are not sufficient to make conclusive statements about the scale issues in pressure-flow scour.
- The effect of various variables on the pressure-flow scour depth has been studied, but there are still many aspects that are debatable (like the effect of sediment size) or have not been explored yet (like the effect of length of contraction).
- Prediction and design equations have been developed for equilibrium scour under upstream clear-water conditions. However, the concept of equilibrium is not well-defined for the pressure-flow scour case. The limited data on temporal evolution of pressure-flow scour suggest that equilibrium is achieved relatively quickly for coarser bed material; however the role of other variables cannot be firmly determined.
- Some prediction equations performed well on training data but not on test data (Tables 4 and 5). This sensitivity toward data sets reduces the applicability of the models. Overall, the models of Umbrell et al. (1998), Lyn (2008), and Shan et al. (2012) have the best prediction performance.
- The coefficients of some models, e.g., those of Umbrell et al. (1998) and Shan et al. (2012), changed significantly during

retraining, raising concerns about their robustness. On contrary, the coefficients of Lyn (2008) did not change during retraining, thus exhibiting robustness against variations in the training data.

- The recommended method for estimating pier scour in pressure-flow conditions in the HEC-18 manual (Richardson and Davis 2001) is to add the pressure-flow scour without a pier with the pier scour in open-channel conditions. However, there is limited empirical evidence to support this recommendation.

Data Availability Statement

All data, models, and code that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

The authors express their sincere thanks to the associate editor and the two anonymous reviewers for their constructive comments and suggestions on the earlier draft of the paper. The support provided by the Science and Engineering Research Board (File No. YSS/2015/000574-SERB) to the second author is also duly acknowledged.

Notation

The following symbols are used in this paper:

- B = width of the channel (m);
- D = pier diameter (m);
- D_{50} = mean sediment size (m);
- G = girder depth (m);
- g = acceleration due to gravity (m/s^2);
- H = difference between approach flow depth, H_a , and contracted passage, H_b (Fig. 1) (m);
- H_a, H_b = approach flow depth and vertically contracted passage (m);
- K_b = vertical blockage to the flow ($H_a - (H_b + w)$) (m);
- L = length of the contraction in the direction of flow (m);
- P_r = discrepancy ratio defined as $(y_{pe}/H_a)_{\text{predicted}} / (y_{pe}/H_a)_{\text{observed}}$;
- q_s = sediment discharge per unit width (m^2/s);
- S_g = specific gravity of sediment;
- t_e = equilibrium time (s);
- t_s = stagnation zone thickness [Fig. 1(b)] (m);
- V_a, V_b = approach flow velocity and average flow velocity inside the contraction (m/s);
- V_c = critical velocity for sediment motion (m/s);
- w = depth of weir overflow (m);
- y_{ope} = maximum equilibrium pier scour in open-channel flow conditions (m);
- y_p, y_{pe} = maximum pressure-flow scour at time t and maximum equilibrium pressure-flow scour depth (m);
- y_{pp} = maximum pier scour in pressure-flow conditions at time t (m);
- y_{ppe} = maximum equilibrium pier scour in pressure-flow conditions (m);
- $(y_{ppe})^c, (y_{ppe})^p$ = vertical contraction scour component and pier scour component of y_{ppe} (Fig. 2) (m);
- ν, ρ = kinematic viscosity of water (m^2/s) and density of water (kg/m^3);

σ = gradation coefficient of sediment size;

T = timescale to nondimensionalize time (s); and

τ_{cr} = critical tractive stress (Pa).

Supplemental Materials

Eqs. (S1)–(S6), Tables S1–S6, Figs. S1–S5, and discussions on the equilibrium time, sensitivity analysis of parameters, and the cross-validation of test results are available online in the ASCE Library (www.ascelibrary.org).

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